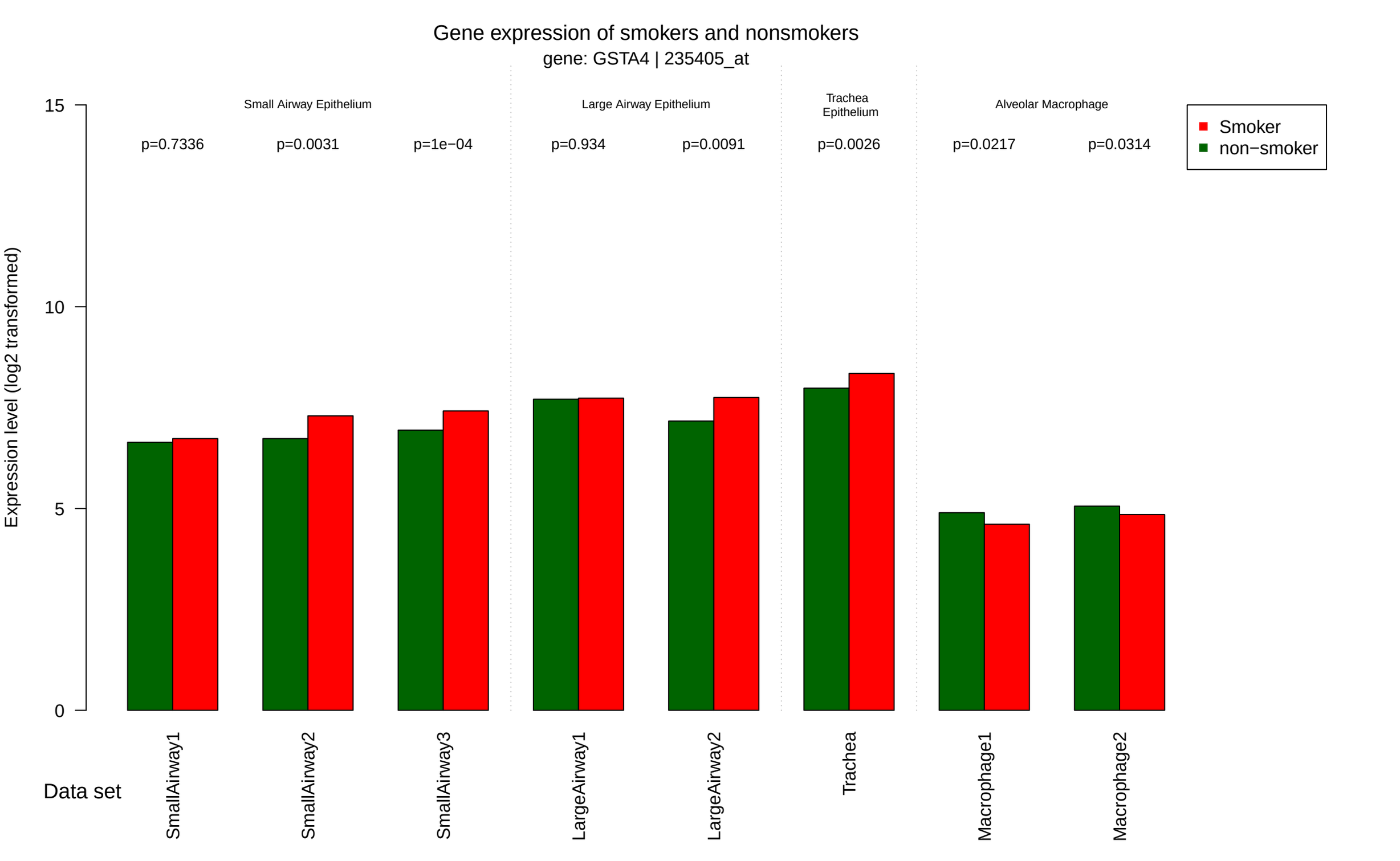
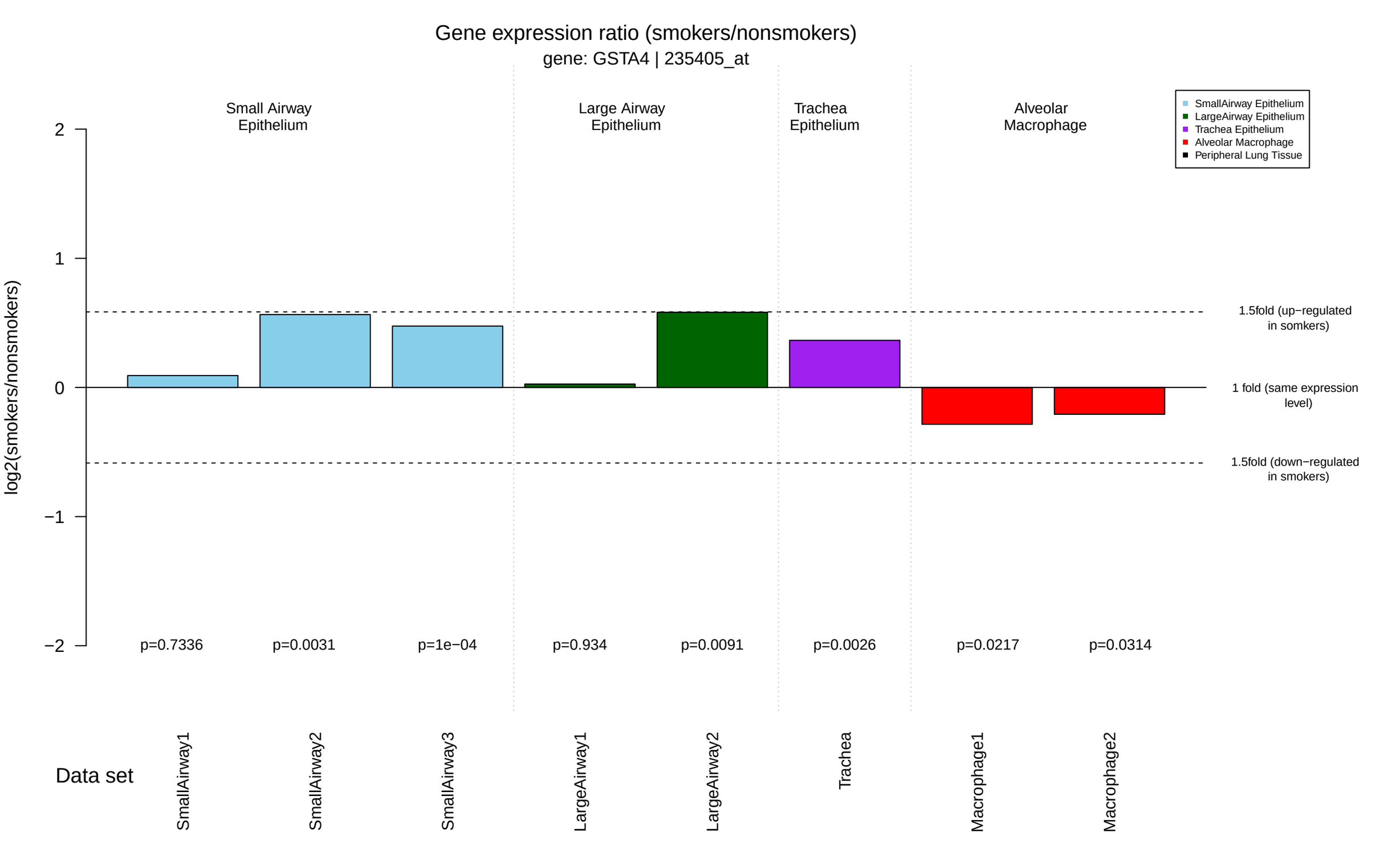
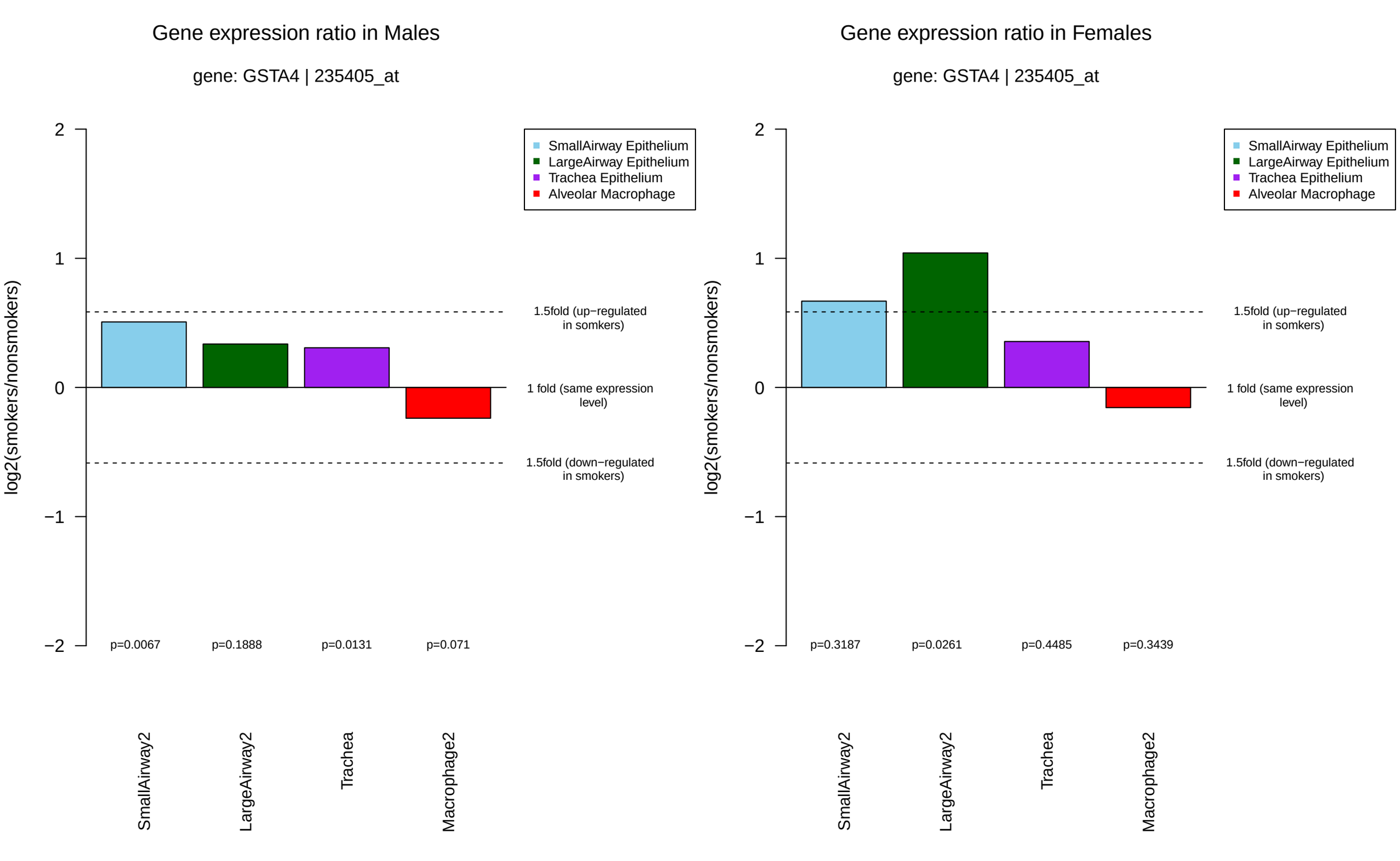
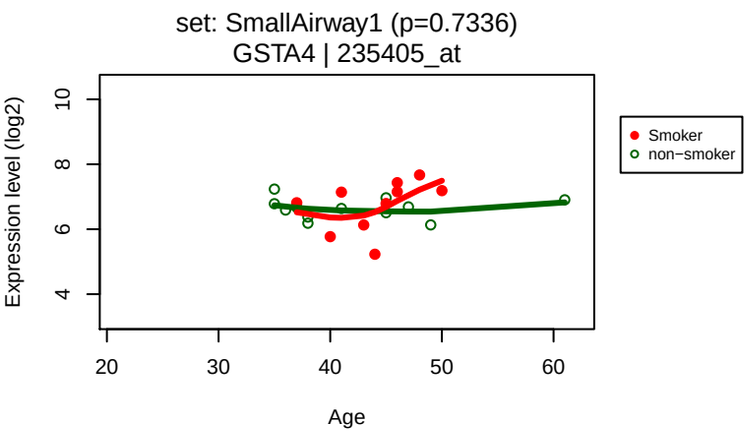
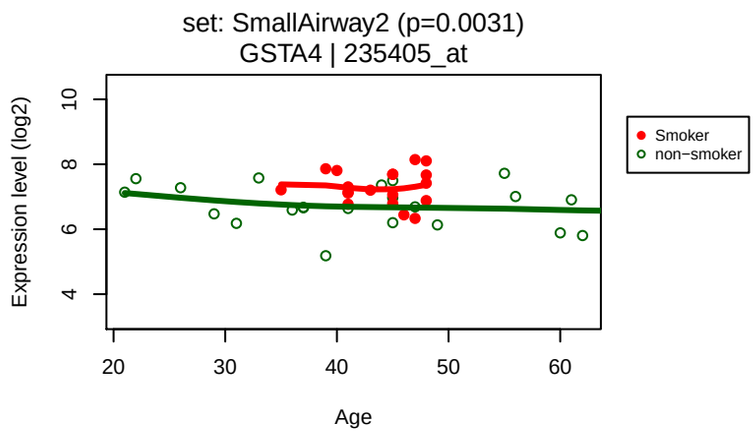
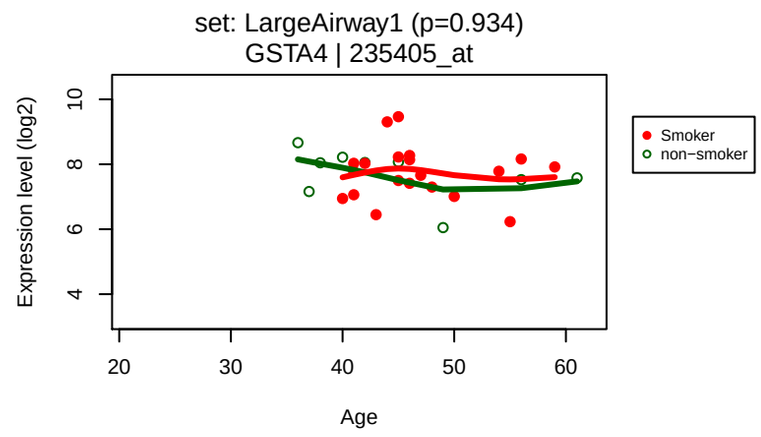

Gene expression of smokers and nonsmokers
gene: GSTA4 | 235405_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7336
p=0.0031
p=1e−04
p=0.934
p=0.0091
p=0.0026
p=0.0217
p=0.0314
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

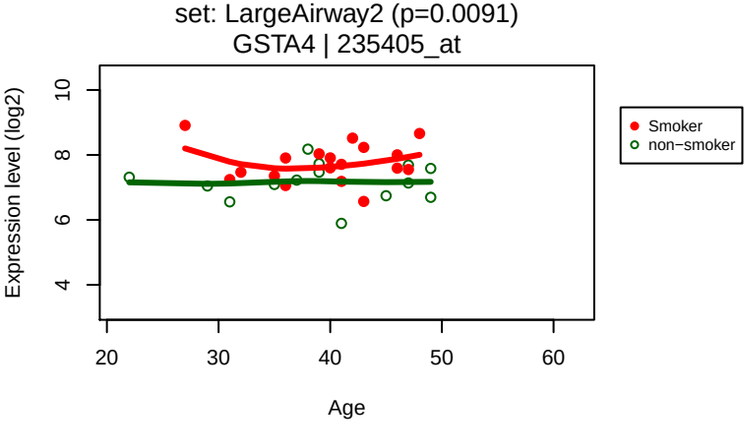

Gene expression ratio (smokers/nonsmokers)
gene: GSTA4 | 235405_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7336
p=0.0031
p=1e−04
p=0.934
p=0.0091
p=0.0026
p=0.0217
p=0.0314
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

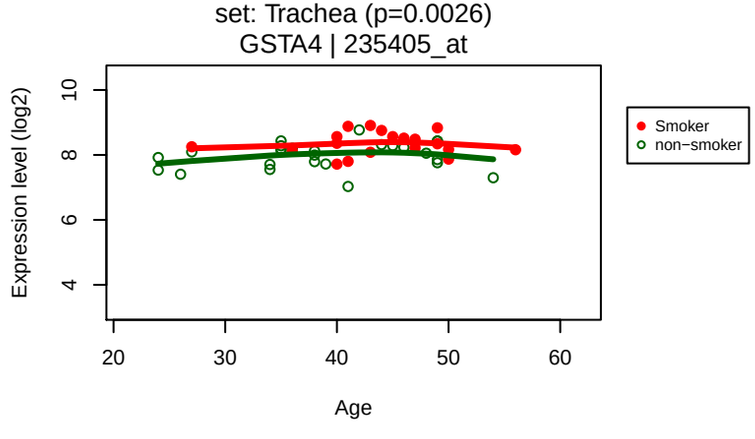

Gene expression ratio in Males
gene: GSTA4 | 235405_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0067
p=0.1888
p=0.0131
p=0.071
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSTA4 | 235405_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3187
p=0.0261
p=0.4485
p=0.3439
SmallAirway2
LargeAirway2
Trachea
Macrophage2

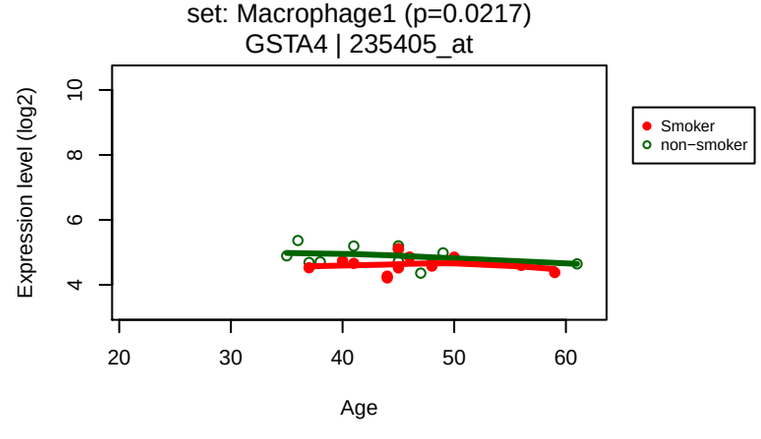

set: SmallAirway1 (p=0.7336)
GSTA4 | 235405_at
Expression level (log2)
Age
Smoker
non−smoker

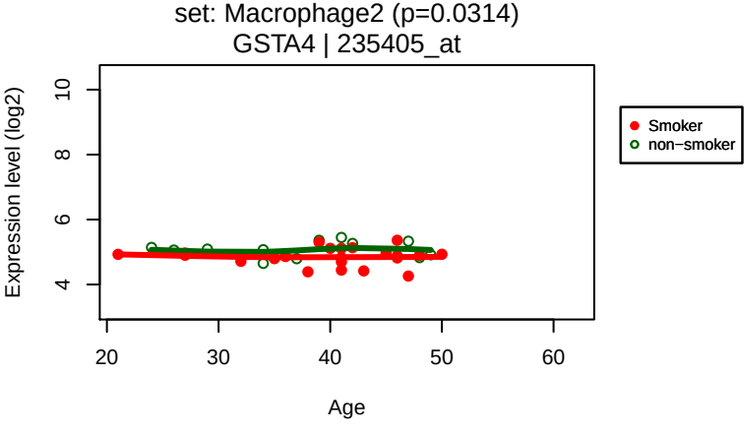

set: SmallAirway2 (p=0.0031)
GSTA4 | 235405_at
Expression level (log2)
Age
Smoker
non−smoker

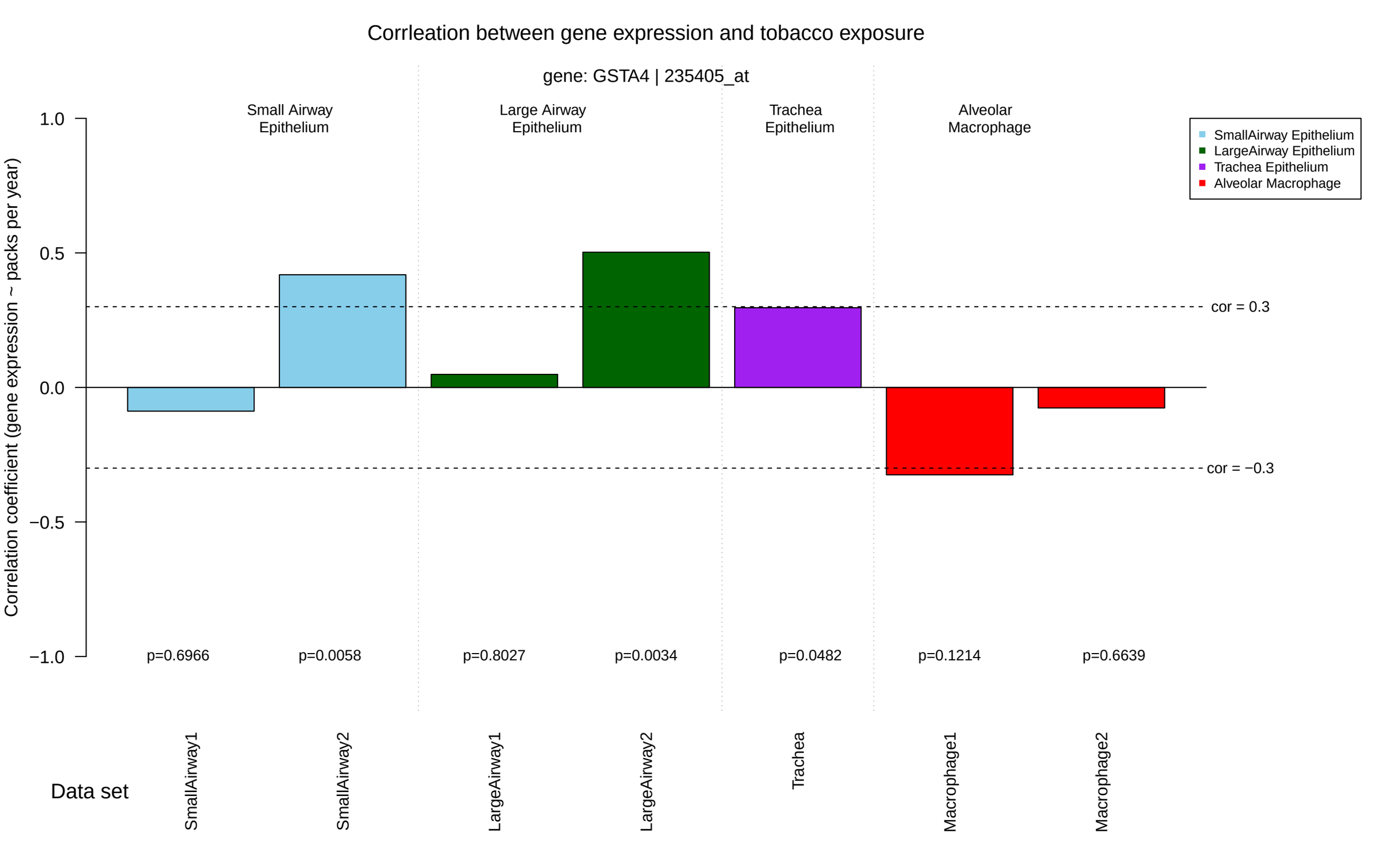

set: LargeAirway1 (p=0.934)
GSTA4 | 235405_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0091)
GSTA4 | 235405_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0026)
GSTA4 | 235405_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0217)
GSTA4 | 235405_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0314)
GSTA4 | 235405_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSTA4 | 235405_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6966
p=0.0058
p=0.8027
p=0.0034
p=0.0482
p=0.1214
p=0.6639
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2